
MULTIDIMENSIONAL PARAMETER DECOMPOSITION FOR MECHANISTIC INTERPRETABILITY

A PREPRINT

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ABSTRACT

Some super cool abstract of all my findings ¹

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¹Very important to add quantization somewhere. <https://arxiv.org/pdf/2303.13506>, new SPD paper in related work and discussion - particularly they also say something about their minimality loss, and lastly extend the transformer thing in the paper to be more clear of the architecture. Particularly no positional encoding?

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1 Introduction

Mechanistic interpretability seeks to reverse-engineer neural networks into a set of human-interpretable mechanisms. Different branches of the field pursue this goal with different focal points: Either through directions in activation space, computational subgraphs, or components of the network’s parameters.

However, both the atomic structure for decomposition and the decomposition itself remain open problems. [Sha+25] Whether a decomposition recovers the computational primitives of a neural network depends on what the ground-truth decomposition is taken to be. With models consisting of independent features, the decomposition is unambiguous, with one atom per feature. If the features are dependent or co-occur, the decomposition becomes underdetermined. Decompositions that split features across multiple atoms, risk losing the structural information, that gives each feature its meaning, while decompositions that group multiple features into a single atom, risk losing granularity by collapsing distinct features into a single unit.

For example, a network computing the volume of a cube can be decomposed into components corresponding to individual side lengths, or into a single component computing the volume directly. The former corresponds to a feature-level decomposition, with one atom per input direction, exposing internal structure but leaving grouping implicit, only recoverable through co-activation patterns. The latter corresponds to a structural-level decomposition, with one atom per computation, exposing the grouping directly but collapsing per side-length information.

Recently, a formal definition of irreducibility was introduced to disambiguate cases such as the cube example.² [Eng+25] Features are irreducible, and thus constitute units of decomposition, if their joint distribution cannot be decomposed into independent factors or into a mixture containing a lower-dimensional component.

²I got this definition wrong, we decompose f not the V ’s, we thus mustn’t show delta orthogonality, and the argument with that the V ’s can be linear combinations isn’t as important anymore. However we still have the same argument, that they can be low dimensional instead of rank 1. Secondly we need to state clearly that f is a function $f : d_{input} \rightarrow \mathbb{R}^{d_{fi}}$

In terms of the decomposition itself, recent work focuses on decomposing parameters, as opposed the activations. These methods aim to recover components that are faithful to the original weights, sparse per input, and individually simple. One such approach, Stochastic Parameter Decomposition (SPD) [BBS25], factorizes networks into rank-one components with a learned per component importance function.

This work extends SPD by examining its behaviour on dependent features, in settings where the multi-dimensional feature hypothesis posits a feature-level or a structural-level unit ³ [Eng+25]. Additionally a reformulation of the importance function is proposed, to encourage that features that are manipulated jointly are grouped together, by computing importance relationally across components.

My findings

2 Related work

2.1 The linear representation hypothesis

A widespread view in mechanistic interpretability is the linear representation hypothesis: high-level concepts are encoded as directions in a network’s representation space [PCV24]. Together with the superposition hypothesis [Elh+22], it forms a central tenet of interpretability: each feature corresponds to a single direction, and a hidden state is a sparse linear combination of feature directions.

Recent work questions how strictly this view should be taken, distinguishing between a weak form of the hypothesis, that some features are linear, from a stronger form that all features are [Men24; Smi24]. Empirically multiple counterexamples to the stronger form have been proposed, include recurrent networks encoding token position by magnitude rather than direction [Cso+24], transformers representing cyclical concepts such as days of the week [Eng+25] or modular arithmetic [Nan+23] in irreducible low-dimensional subspaces.

2.2 Activation-space decomposition

Decomposing a network’s activations predominantly focuses on disentangling layer activations into interpretable features. Here, sparse autoencoders (SAE) are one of the most widely researched methods [Cun+23b; Bri+23], that learn an overcomplete dictionary that expresses each hidden state as a sparse combination of one-dimensional features. However, these features are not necessarily aligned with the network’s computation, due to feature absorption, inconsistencies across dictionary sizes, and a failure to capture feature geometry. [Cha+25; Lea+25; Men24].

Circuit discovery extends feature analysis to identify how a computation is processed by the network’s structure, by identifying minimal subgraphs required to perform a given task, without loss of performance. [Ola+20; Mar+25]. However, these methods generally do not account for the polysemantic and densely connected structure of neural networks, where the functional behaviour can be attributed to multiple distinct but functionally equivalent subgraphs [Mél+25].

2.3 Parameter-space decomposition

Recently a newer line of work aims to decompose the network’s parameters directly, instead of the activations. Transcoders [DCN24] represent an intermediate step that learns a sparse approximation of an MLP layer. Parameter decomposition methods go further, factorising the network’s weights directly, with the goal of recovering the network’s computational primitives.

Most directly relevant to this work is the description-length-driven line of Attribution-based Parameter Decomposition [Bra+25] and Stochastic Parameter Decomposition (SPD) [BBS25], which has recently been explored in small transformer models [CR25] and cross-layer clustering [Ses+26]. While APD decomposes a network into multidimensional components, the method is sensitive to hyperparameters, and does not scale beyond toy settings due to computational complexity. SPD addresses this by decomposing each weight matrix into rank-one subcomponents and learning a per-input importance function over them, at the cost of requiring post-hoc clustering to recover higher-rank mechanisms; both methods are detailed in Section 3.1. Other parameter-space approaches include local loss landscape decomposition [CBS25].

³I mean feature-level, structure level is wrong under this view but it sounds good

3 Method

3.1 Background on SPD

Attribution-based Parameter Decomposition (APD) [Bra+25] and Stochastic Parameter Decomposition (SPD) [BBS25] both aim to decompose a network’s parameters into a set of components that are faithful to the original weights, minimal in the number required per input, and individually as simple as possible.

To formalise this, consider a neural network $f(x, W)$ with weight matrices $W = \{W^1, \dots, W^L\}$. A set of parameter components $\{W_c^l\}_{c=1}^C$ are said to comprise the network’s mechanisms if they satisfy the following properties:

- **Faithfulness:** The parameter components should sum to the parameters of the original network.
- **Minimality:** As few parameter components as possible should be required to replicate the network’s output on any given input.
- **Simplicity:** Each component should span as few weight matrices and as few ranks as possible.

APD [Bra+25] approaches this by decomposing the network into a set of parameter components without constraining their rank, using a gradient-based attribution that selects the top- k components most responsible for any given input. While effective on small models, APD scales poorly: its memory cost grows with the number of components, as each component has the same dimensionality as the original network, and its top- k attribution is sensitive to hyperparameter choice.

SPD [BBS25] addresses these limitations by enforcing a rank-one structure on each component, allowing more components to be learned at fixed memory cost, and replacing top- k attribution with a learned causal importance function. Formally, each weight matrix W^l is decomposed into a sum of C rank-one subcomponents:

$$W_{i,j}^l \approx \sum_{c=1}^C U_{c,i}^l V_{c,j}^l, \quad \text{where } U^l \in \mathbb{R}^{C \times d_{\text{out}}}, V^l \in \mathbb{R}^{C \times d_{\text{in}}} \quad (1)$$

where C is a hyperparameter, for the number of components for each layer, l indexes the layer and i, j are the hidden indices. The subcomponent count, C , may be chosen to be larger than both d_{in} and d_{out} , enabling the method to learn features in superposition [BBS25].

To enforce the properties above, three losses are introduced, which are optimised jointly. Faithfulness is enforced through a mean squared error loss between the original network’s weights and the the sum of subcomponents:

$$\mathcal{L}_{\text{faithfulness}} = \frac{1}{N} \sum_{l=1}^L \sum_{i,j} \left(W_{i,j}^l - \sum_{c=1}^C U_{i,c}^l V_{c,j}^l \right)^2 \quad (2)$$

where N is the total parameter count in the target model.

Optimization for simplicity and minimality is achieved through a stochastic masking procedure. Intuitively, if a subcomponent is not necessary for reconstruction, it can be sampled to zero without loss of faithfulness. A causal importance function $\Gamma : \mathcal{X} \rightarrow [0, 1]^{C \times L}$ is trained to learn the lower bound of the sampling distribution. In SPD, Γ is a set of independent MLPs γ_c^l that each take a layer’s activation and output a scalar importance for the corresponding subcomponent:

$$g_c^l(x) = \Gamma_c^l(x) = \sigma_H(\gamma_c^l(h_c^l(x))), \quad h_c^l(x) = \vec{V}_{c,\cdot}^{l\top} \vec{a}^l(x) \quad (3)$$

where σ_H is a hard sigmoid and $\vec{a}^l(x)$ is the activation at layer l . These importance values serve as the lower bound for how much a subcomponent can be masked. With $r_c^l \sim \mathcal{U}(0, 1)$ sampled per subcomponent, the masked weight matrix is given by:

$$m_c^l(x, r) = g_c^l(x) + (1 - g_c^l(x)) r_c^l, \\ W^l(x, r) = \sum_{c=1}^C \vec{U}_{i,\cdot}^l m_c^l(x, r) \vec{V}_{c,j}^l, \quad (4)$$

These masked matrices are trained to reconstruct the original network’s output.

$$f(x | \{W^l(x, r)\}_{l=1}^L) \approx f(x | \{W^l\}_{l=1}^L). \quad (5)$$

This is done through both a layer-wise and a global reconstruction loss. The layer-wise version is necessary to provide a stronger training signal as randomly masking all components at the same time can be too noisy to learn from.

$$\begin{aligned}\mathcal{L}_{\text{stochastic-recon}} &= \frac{1}{S} \sum_{s=1}^S D(f(x | W'(x, r^s)), f(x | W)), \\ \mathcal{L}_{\text{stochastic-recon-layer}} &= \frac{1}{LS} \sum_{l=1}^L \sum_{s=1}^S D(f(x | W^1, \dots, W'^l(x, r^{(s)}), \dots, W^L), f(x | W)).\end{aligned}\tag{6}$$

where D is a task-specific divergence measure, such as MSE or KL divergence.

Lastly, to enforce minimality, the importance function is trained to activate only the subcomponents needed for reconstruction. The loss balances reconstruction accuracy against the number of active subcomponents, creating a tradeoff between faithfulness and minimality. The goal is to activate only the subcomponents required to reproduce the target model’s output while leaving the rest inactive.

$$\mathcal{L}_{\text{minimality}} = \sum_{l=1}^L \sum_{c=1}^C (g_c^l(x))^p, \quad p > 0.\tag{7}$$

The full SPD objective is a weighted sum of these terms:

$$\mathcal{L}_{\text{SPD}} = \mathcal{L}_{\text{faithfulness}} + \beta_1 \mathcal{L}_{\text{stochastic-recon}} + \beta_2 \mathcal{L}_{\text{stochastic-recon-layer}} + \beta_3 \mathcal{L}_{\text{minimality}}.\tag{8}$$

3.2 Irreducibility and the natural unit of decomposition

To appropriately evaluate the performance of a multi-dimensional feature decomposition method, it’s necessary to have a clear definition of what the ground-truth decomposition is. The original superposition hypothesis states that the hidden states can be represented as a linear combination of sparse one-dimensional features. [Elh+22; Eng+25]

Definition: The hidden state $x_{i,l}$ can be written as:

$$x_{i,l} = \sum_i v_i f_i(t)\tag{9}$$

where $v_i \in \mathbb{R}^d$ are pairwise δ -orthogonal directions ⁴ and $f_i(t) \in \mathbb{R}$ are sparse scalar activations, with $f_i(t) = 0$ outside the domain of feature i .

Recent work has shown, that this may be insufficiently strict [Smi24; Eng+25], as it does not capture the structure of multi-dimensional features. In particular, transformer models are shown to organise cyclical concepts in low-dimensional subspaces that do not decompose into meaningful linear components, but are instead manipulated jointly [Eng+25].

To accomodate these features, the same paper proposes a correction to the superposition hypothesis, alongside a formal definition of irreducibility that defines, when a feature is in its atomic form and when it can be decomposed into smaller units without loss of information.

Definition: A feature $f : \mathbb{R}^i \rightarrow \mathbb{R}^{d_f}$ is truly multi-dimensional, if it cannot be written as a sum of two independent features or as a the sum of two non-co-occurring features. More formally, a feature is reducible into features a and b if there exists an affine transformation

$$f \mapsto Rf + c \equiv \begin{pmatrix} a \\ b \end{pmatrix}\tag{10}$$

for some orthonormal $d_f \times d_f$ matrix R , and additive constant c , such that the joint distribution $p(a, b)$ satisfies one of two conditions:

- **p is separable:** $p(a, b) = p(a) p(b)$
- **p is a mixture of two disjoint distributions:** $p(a, b) = w p(a) \delta(b) + (1 - w) p'(a, b|b \neq 0)$,

Here a and b are the first k and remaining $d_f - k$ components of $Rf + c$, p is conditioned on inputs where f is active, $w \in (0, 1)$ and δ is the Dirac delta ⁵. In the mixture, the two distributions have disjoint support, with one distribution having no mass where b is active, and the other having no mass where b is zero.

⁴should i say what delta is or is it clear that its small?

⁵can i get away with just writing $p(a|b = 0)$

Under this decomposition, the superposition hypothesis takes the form:

Definition: Every hidden state $x_{i,l}(t)$ can be written as a sum of feature contributions

$$x_{i,l}(t) = \sum_i V_i f_i(t) \quad (11)$$

where each $f_i(t) \in \mathbb{R}^{d_{f_i}}$ is an irreducible feature with internal dimensionality $d_{f_i} \geq 1$, each $V_i \in \mathbb{R}^{d \times d_{f_i}}$ and $\|V_i^\top V_j\| < \delta$ for $i \neq j$.

Secondly, and more importantly for SPD, if the multidimensional superposition hypothesis holds, some features f may require multiple rank-one subcomponents. It is not trivial that an irreducible function f (e.g. as shown in Section ??) arises from sampling two dependent and non-overlapping features that can still be compressed into a single rank-one component if the model learns to group them along a shared direction.

However, when the model learns genuinely multi-dimensional features, these cannot in general be represented by a rank-one subcomponent. Instead, they occupy structured higher-dimensional subspaces, implying either the need for higher-rank representations or post hoc clustering of the underlying rank-one components.

3.3 Rank- k subcomponents and relational importance

In the original SPD paper, the subcomponents are constructed as rank-one matrices, through an outer product of two vectors. This formulation is however incompatible with the multi-dimensional superposition hypothesis. (Section 3.2) A decomposition into lower-dimensional subcomponents, necessarily forces features to be split across multiple subcomponents, or to be approximated through a single subcomponent at the cost of reconstruction.

To represent multi-dimensional features, the pairs of vectors are replaced with pairs of matrices.

$$W_{i,j}^l \approx \sum_{c=1}^C \sum_{r=1}^k U_{c,i,r}^l V_{c,j,r}^l, \quad \text{where } U^l \in \mathbb{R}^{C \times d_{\text{out}} \times k}, V^l \in \mathbb{R}^{C \times d_{\text{in}} \times k} \quad (12)$$

Each subcomponent $U_c^l V_c^{l\top} \in \mathbb{R}^{d_{\text{out}} \times d_{\text{in}}}$ has rank at most k , recovering the original SPD definition when $k = 1$. As in the multi-dimensional superposition hypothesis, k is an upper bound on each subcomponent’s representational capacity; the effective rank may be lower,

In SPD, each subcomponent’s importance is computed independently through an MLP followed by a sigmoid non-linearity, allowing multiple subcomponents to be active for a given input and thus distribute a feature across them, even when it is irreducible. Likewise, the minimality loss does not translate to the rank- k setting, where each subcomponent can represent multiple feature directions: it imposes no penalty for distributing a feature across multiple subcomponents, nor for a single subcomponent to encode parts of multiple features. (Section 3.1)

To address this, the causal importance function is redefined to maximize the conditional probability, that a given subcomponent W_c^l , is responsible for a given feature, conditioned on the activations of all other subcomponents in the same layer.

$$g_c^l(x) = p(W_c^l \mid \{h_{c'}^l(x)\}_{c'=1}^C), \quad h_c^l(x) = V_c^{l\top} \bar{a}^l(x) \in \mathbb{R}^k \quad (13)$$

where $\bar{a}^l(x)$ is the activation of layer l and C is the set of subcomponents in layer l . The distribution is computed with a classic transformer block operating over subcomponents, with a softmax over their activations. [Vas+23]

This produces a normalized assignment over subcomponents, $\sum_{c=1}^C g_c^l(x) = 1$, causing the original minimality loss to become degenerate, as each layer contributes a constant value independent of the model state. Instead, it’s replaced with entropy, to pressure the assignment to concentrate on as few subcomponents as possible.

$$\mathcal{L}_{\text{minimality}} = -\frac{1}{|L|} \sum_{l=1}^L \sum_{c=1}^C g_c^l(x) \log g_c^l(x) \quad (14)$$

An unfortunate limitation of this formulation is its computational cost. Computing relational importance via a transformer over subcomponents scales quadratically with the number of subcomponents compared to the linear cost of the original independent importance functions.⁶

⁶Play with attention approximations

4 Results

4.1 Overview of experiments

To evaluate how the proposed changes affect SPD’s performance, and how these relate to the multi-dimensional superposition hypothesis, the result section is structured around three experiments ⁷, each with the purpose of isolating a specific structural property. For an overview see table 1.

The first experiment reuses the Toy model of Superposition (TMS) [Elh+22] once sampled with independent features and once with paired features, where these features are sampled in pairs, creating the simplest case of statistical dependence. In particular, the model learns to group the paired features along a single direction, allowing rank 1 subcomponents to capture full features, without having to split them across multiple subcomponents.

The second experiment also reuses a known experiment from SPD, the Toy model of compressed computation, where a model must learn to compute more functions, than the number of available directions in a lower dimensional space. Particularly, this experiment is interesting to explore how softmax and attention affect the decomposition. Softmax in particular, is best when the ground truth decomposition is meant to pick a singular subcomponent for each feature, due to its properties[some source], the toy model of compressed computation, does not allow for a one-hot assignment, as no disentanglement is possible, and to stay faithful to the original model, should learn to distribute mass across multiple subcomponents.

The third experiment generalizes the paired setting to a k-dimensional case by computing the volume of a simplex from its vertices. The ground truth decomposition is still shown to be dependent, but the features can now not be contained in a single rank-one subcomponent, as the geometry of the feature is more complex. This experiment serves to test how well softmax and sigmoid gating can group feature parts together, when their expressiveness is increased.

	Toy Model of Superposition	Toy Model of Dependent Superposition	Toy Model of Compressed Computation	Toy Model of Geometry
Objective	$x \in \mathbb{R} \mapsto x$	$x \in \mathbb{R}^d \mapsto x$	$x \in \mathbb{R} \mapsto x + \text{ReLU}(x)$	$x \in \mathbb{R}^d \mapsto \text{vol}(x) \in \mathbb{R}^+$
Statistical dependence	×	✓	×	✓
Formal feature dimension	1	2	1	$n(n+1)$
Component rank required	1	1	1	k
Ground-truth decomposition	One-hot	One-hot per pair	Distributed	One-hot per simplex

Table 1: Overview of the toy models and their properties.

4.2 Toy Model of Superposition

The Toy Model of Superposition (TMS) [Elh+22] is trained to reconstruct sparse feature activations through a lower-dimensional bottleneck that forces the model to learn features in superposition. The architecture is a single-hidden-layer autoencoder with tied weights:

$$\hat{x} = \text{ReLU}(W^\top Wx + b), \quad W \in \mathbb{R}^{d_{\text{hidden}} \times d_{\text{features}}}, \quad d_{\text{features}} \gg d_{\text{hidden}}. \quad (15)$$

TMS is studied in two regimes: one with independent features, reproducing the original SPD setup, and one with paired features that introduces statistical dependence. For both regimes, the rank of each subcomponent is kept at 1, both because this is the ground-truth structure and to isolate the effect of softmax-and-attention.

To quantify how well the learned subcomponents recover the target model, two metrics are computed across both regimes. The mean max cosine similarity (MMCS) [Cun+23a] measures directional alignment between each target

⁷maybe 4

feature and its best-matching learned subcomponent:

$$\text{MMCS} = \frac{1}{d_{\text{features}}} \sum_{j=1}^{d_{\text{features}}} \max_c \left(\frac{\vec{U}_{:,c} V_{c,j} \cdot W_{:,j}}{\|\vec{U}_{:,c} V_{c,j}\|_2 \|W_{:,j}\|_2} \right). \quad (16)$$

where $c \in C$ is the parameter component index and $j \in \{1, \dots, d_{\text{features}}\}$ is the input feature index. A value of 1 indicates that every target feature has a perfectly aligned subcomponent.

Apart from directional alignment, the strength of the alignment is also measured, to detect feature shrinkage, a problem apparent in both Sparse Autoencoders and Attribution-Based Parameter Decomposition [JT24; WS24; Bra+25]. The mean L_2 ratio (ML2R) compares the euclidean norm of the best-aligned subcomponent to the target feature [BBS25]:

$$\text{ML2R} = \frac{1}{d_{\text{features}}} \sum_{j=1}^{d_{\text{features}}} \frac{\|\vec{U}_{:, \text{mcs}(j)} \vec{V}_{\text{mcs}(j), j}\|_2}{\|W_{:, j}\|_2} \quad (17)$$

where $\text{mcs}(j)$ is the index of the most aligned subcomponent for feature j . A ratio near 1 indicates the feature is fully captured by its best-aligned subcomponent.

4.2.1 Independent features

Each feature is drawn independently from a uniform distribution over $[0, 1]$. As each feature is one-dimensional and independent of the others, the ground-truth decomposition is a set of d_{features} rank-one matrices, each non-zero only along the c -th column $W_{:,c}$ of the target weights.

Like in the original SPD paper, two TMS models are trained: TMS₅₋₂ with $d_{\text{features}} = 5$ and $d_{\text{hidden}} = 2$, and TMS₄₀₋₁₀ with $d_{\text{features}} = 40$ and $d_{\text{hidden}} = 10$ with an additional identity layer.

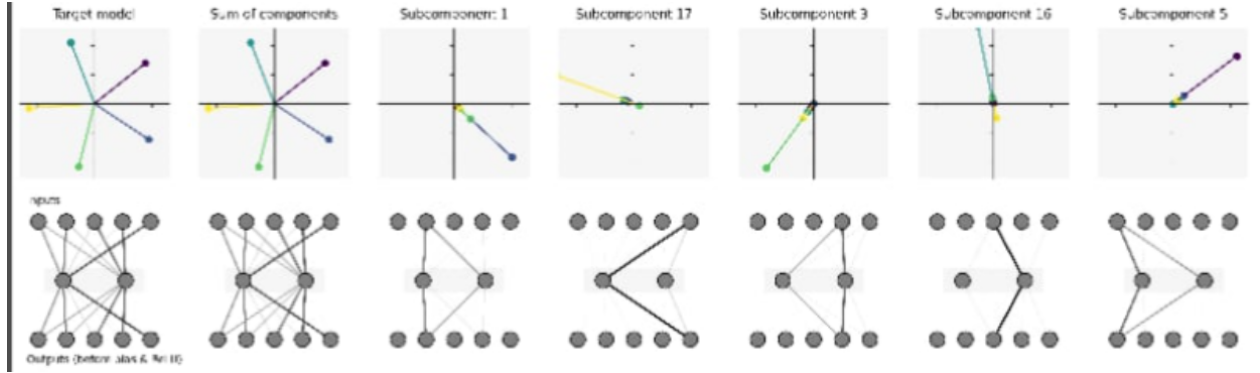


Figure 1: Decomposition of TMS₅₋₂ using Softmax. The first two panels show the target model’s weight columns and the sum of learned subcomponents. Subsequent panels show the individual subcomponents, ordered by decreasing norm. Although 20 subcomponents are allocated, only the first 5 carry meaningful norm and contribute to reconstruction, while the remaining 15 have negligible weight and are effectively pruned by the method. Each active subcomponent recovers primarily one feature direction, with minor leakage across components.

SPD achieves near perfect alignment on both models, with MMCS and ML2R approximately equal to 1. The softmax and attention variant, while achieving near-zero losses, does not manage to perfectly align its components with the target model nor concentrate the entire feature magnitude within a single subcomponent, resulting in a lower ML2R with high variance. For the larger model this skews the geometry further.

Insert a table for this!⁸

While the softmax variant does not recover the exact geometry, it correctly concentrates the entire magnitude on the right number of subcomponents, with the rest pruned to near-zero (Figure 1). In terms of causal importance values, the softmax and original SPD method show the same one-hot assignment for each of the five features in TMS₅₋₂, with some degradation in the larger model where certain subcomponents are causally important for multiple features.

⁸important

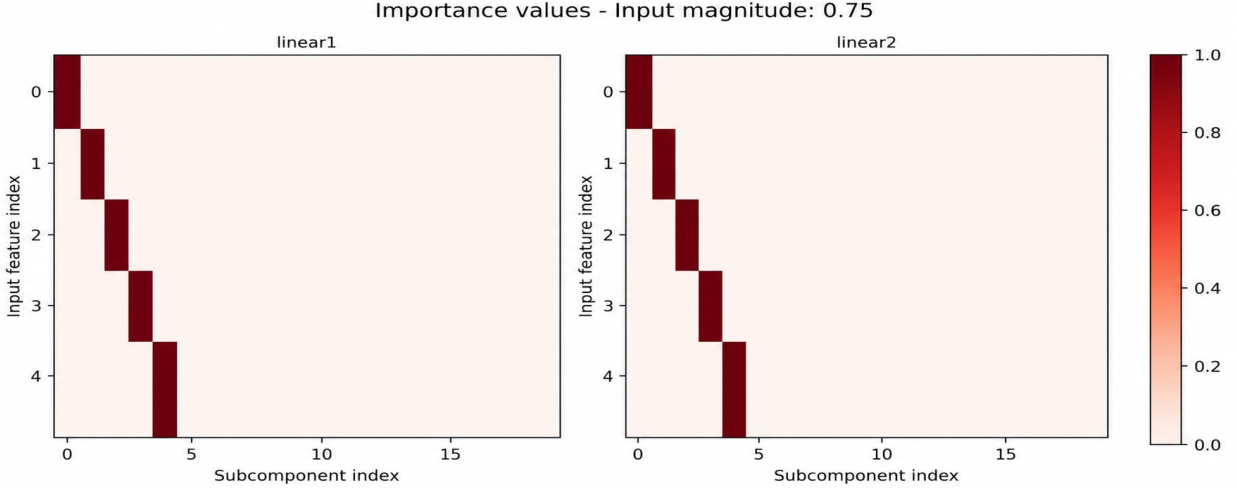


Figure 2: Causal importance values for the softmax variant on TMS_{5-2} , showing a one-hot assignment for each of the five features.

4.2.2 Paired features

In this regime, features are sampled in pairs to create the simplest form of dependence. Features are partitioned into pairs $\{f_{2i-1}, f_{2i}\}_{i=1}^{20}$, with each pair zero with probability $1-p$, and otherwise each pair member drawn independently from $\mathcal{U}(\epsilon, 1)$. The setup uses $d_{\text{features}} = 40$ and $d_{\text{hidden}} = 10$.

Each input is 2-dimensional at construction, but the trained model learns a feature mapping that aligns both pair members along a single direction, with the two features in each pair being manipulated jointly. (Table 2), with within-group cosine similarity averaging 0.999 and across-group similarity averaging 0.053. The pair’s hidden representation therefore collapses to a 1D subspace. Due to this lower-dimensional mapping, paired features can be captured with a single rank-1 subcomponent, without needing to split them across multiple subcomponents.

Group	Mean	Std Dev	Range
Within-group	0.999	0.010	[0.997, 1.000]
Across-group	0.053	0.223	[-1.000, 0.038]

Table 2: Cosine similarity between encoder columns $W_{:,i}$ and $W_{:,j}$, partitioned by group membership.

The ground-truth decomposition is therefore 20 rank-1 subcomponents, with feature indices $\{2i-1, 2i\}$ assigned to subcomponent i , where the causal importance function should assign all importance to the correct subcomponent for each feature pair, and zero importance to the rest.

In terms of losses, both SPD and Softmax achieve total losses around 10^{-6} , with no significant differences between the two methods. In terms of MMCS, both methods achieve near-perfect alignment, with values around 0.999.

There needs to be another table here.

However, the softmax variant shows a much lower ML2R with high variance, indicating that it struggles to capture the full magnitude of the features within single subcomponents. In terms of causal importances, the two methods show very different patterns, despite identical losses and MMCS. SPD correctly identifies that features should be paired in single subcomponents but does not isolate each pair into its own subcomponent, with several subcomponents carrying importance for multiple feature pairs. The softmax variation correctly assigns each feature pair to a single subcomponent, with the exception of one pair (Figure 3).

Overall across the three settings, SPD with sigmoid gating produces better results on both MMCS and ML2R, despite the more entangled importance assignments observed in the paired regime (Figure 3). The MMCS gap is generally small, but with substantially larger variation in ML2R.

Referring back to the TMS_{5-2} decomposition shown in Figure 1, the softmax method appears to recover a slightly rotated geometry, with residual feature magnitude distributed across adjacent features. The geometric offset may arise

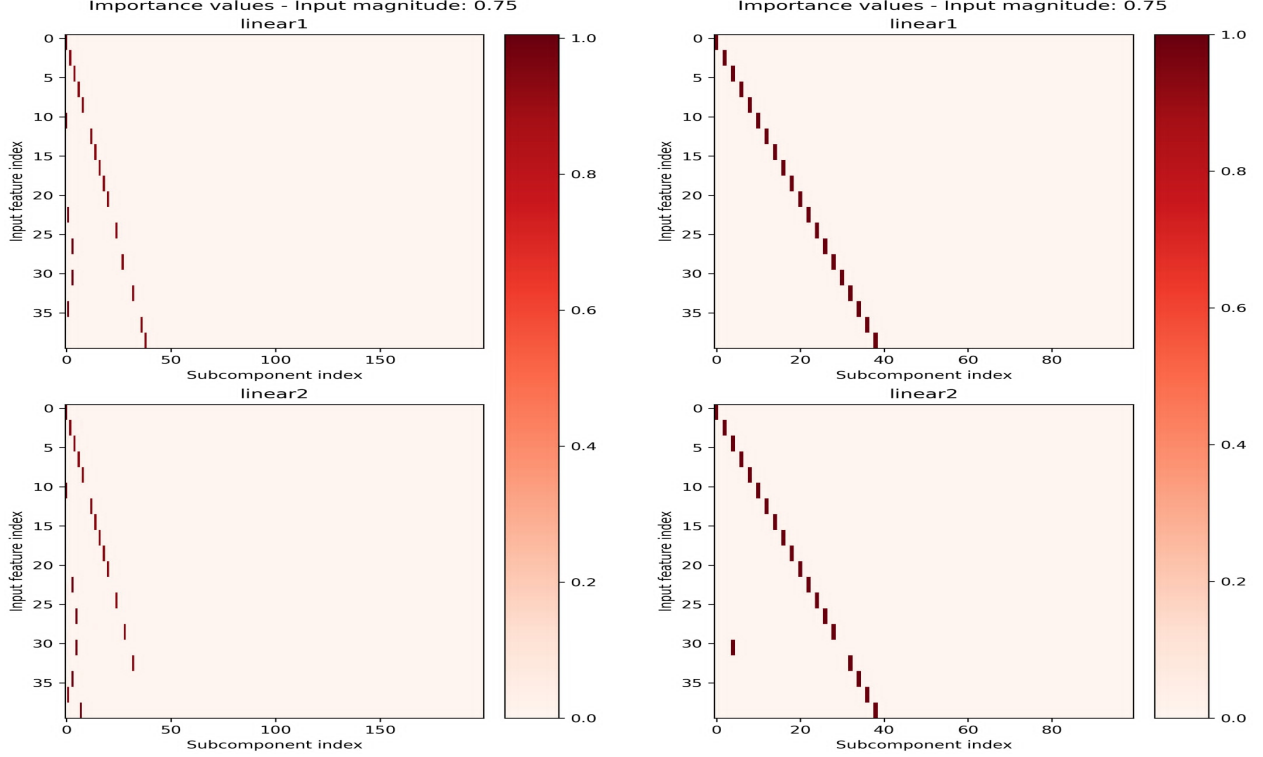


Figure 3: Left: SPD decomposition of the paired TMS model. Right: Softmax of the paired TMS model. SPD finds the correct structure of pairing features together but does not attribute one feature pair per subcomponent. Softmax achieves the desired importance decomposition, with the exception of one feature pair. SPD was trained with 200 subcomponents and Softmax with 100, as these yielded the best results for each method.

	TMS ₅₋₂		TMS ₄₀₋₁₀		TMS ₄₀₋₁₀ paired	
	SPD	Softmax	SPD	Softmax	SPD	Softmax
MMCS	1.00	0.985 ± 0.016	1.00	0.812 ± 0.105	0.9997	0.9920 ± 0.003
ML2R	1.00	0.797 ± 0.205	1.055 ± 0.045	0.623 ± 0.231	0.796 ± 0.434	0.732 ± 0.350

Table 3: Decomposition quality across both TMS regimes. SPD refers to the original method [BBS25]; Softmax refers to the relational importance function of Section 3.3. If the standard deviation is not reported, it’s because its smaller than 0.001.

as a compensatory mechanism for this residual leakage, allowing the combined subcomponents to maintain accurate reconstruction despite imperfect feature isolation.

4.3 Toy Model of Compressed Computation

The Toy Model of Compressed Computation, a model that was already studied by SPD [BBS25], APD [Bra+25] and more recently by [Bha+25], is a toy model trained to compute the elementwise function $y_i = x_i + \text{ReLU}(x_i)$ through a residual MLP with a lower-dimensional hidden layer.

The architecture consists of an embedding matrix W_E , an unembedding matrix $W_U = W_E^\top$, and a residual MLP with input projection $W_{\text{in}} \in \mathbb{R}^{50 \times d_{\text{emb}}}$ and output projection $W_{\text{out}} \in \mathbb{R}^{d_{\text{emb}} \times 50}$. The input vectors are sparse activations sampled from a uniform distribution over $[-1, 1]$, with the model trained to minimize the MSE between the output and the target function. The model’s computation can be expressed as:

$$y = W_U(W_E x + W_{\text{out}}(\text{ReLU}(W_{\text{in}} W_E x))).$$

For comparability with previous work, the experiments use $d_{\text{features}} = 100$, $d_{\text{MLP}} = 50$, rank $k = 1$ matrices, $C = 200$ subcomponents and lastly an embedding dimension $d_{\text{emb}} = 1000$.

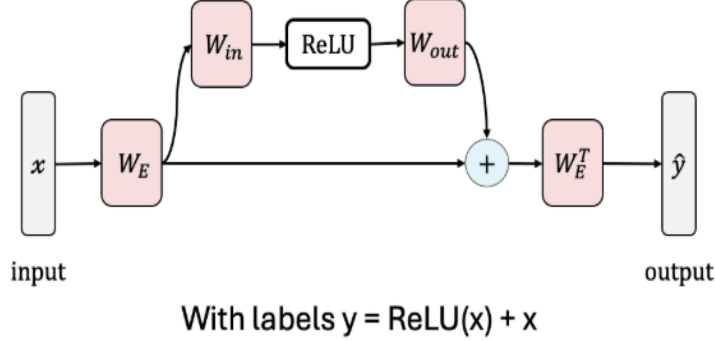


Figure 4: Architecture of the Toy Model of Compressed Computation. Picture taken from [Bha+25]

A naive decomposition of this model would be to let each of the neurons in the hidden layer compute one of the 100 ReLU computations. However this would perform poorly, on the remaining 50 neurons. Instead, to achieve a better reconstruction, the model seems to learn a more efficient representation by assigning multiple features to the same neuron.

The ground truth for W_{in} is clear: each subcomponent should be causally important for one input feature matching the independent one-dimensional structure of the inputs.

For W_{out} , the ground truth is less clear. APD [Bra+25] initially suspected a monosemantic decomposition where each subcomponent corresponds to one output feature, to be the most truthful decomposition. SPD [BBS25] on the other hand reports a polysemantic decomposition in which 50 subcomponents each contribute to multiple features while the remaining are pruned to near-zero norm.

Another paper [Bha+25] supports the interpretation of SPD, showing that the model does not implement 100 separate computations through superposition but rather that they jointly operate within a shared 50-dimensional subspace.

4.3.1 Sampling effect on loss

Before analyzing the decomposition, this experiment will also be used to compare the effect of different sampling distributions on the loss and decomposition quality. Here the two methods SPD and SPD with softmax are compared across two sampling regimes: one-hot sampling, where only one feature is active at a time, and at-least-zero sampling, where each input is independently sampled with probability p .

Softmax generally produces lower losses when sampling is done with one-hot inputs rather than sampling every feature independently. Most noticeably, the Minimality loss, does not converge under at-least-zero sampling. This is unsurprising, with how the loss is defined to maximize the probability of a feature belonging to a single subcomponent. However if the input distribution is not one-hot, the input is a noisy and potentially meaningless combination of unrelated features.

Sigmoid-based importance does not exhibit a significant difference between the two sampling methods, likely since each subcomponent’s importance is evaluated independently, and the loss measures the magnitude of subcomponent activation (Section 3.1) to the correct contribution.

Metric	SPD		Softmax	
	One-hot	At-least-zero	One-hot	At-least-zero
Faithfulness	1.5×10^{-6}	1.4×10^{-6}	3×10^{-5}	5×10^{-5}
Stochastic reconstruction	1.8×10^{-4}	1×10^{-5}	9×10^{-5}	1×10^{-4}
Layerwise stochastic reconstruction	1×10^{-5}	1×10^{-5}	7×10^{-5}	9×10^{-5}
Output reconstruction	3×10^{-6}	4×10^{-6}	8×10^{-5}	1.6×10^{-4}
Minimality	3.83	3.95	8×10^{-5}	4.06

Table 4: Reconstruction quality across sampling regimes for the Residual MLP.

4.3.2 Causal importance

The experimentation with Softmax seems to replicate a similar structure to that found in APD [Bra+25], where one subcomponent is causally important for one feature, while the remaining subcomponents have negligible importance for that feature in both layers.

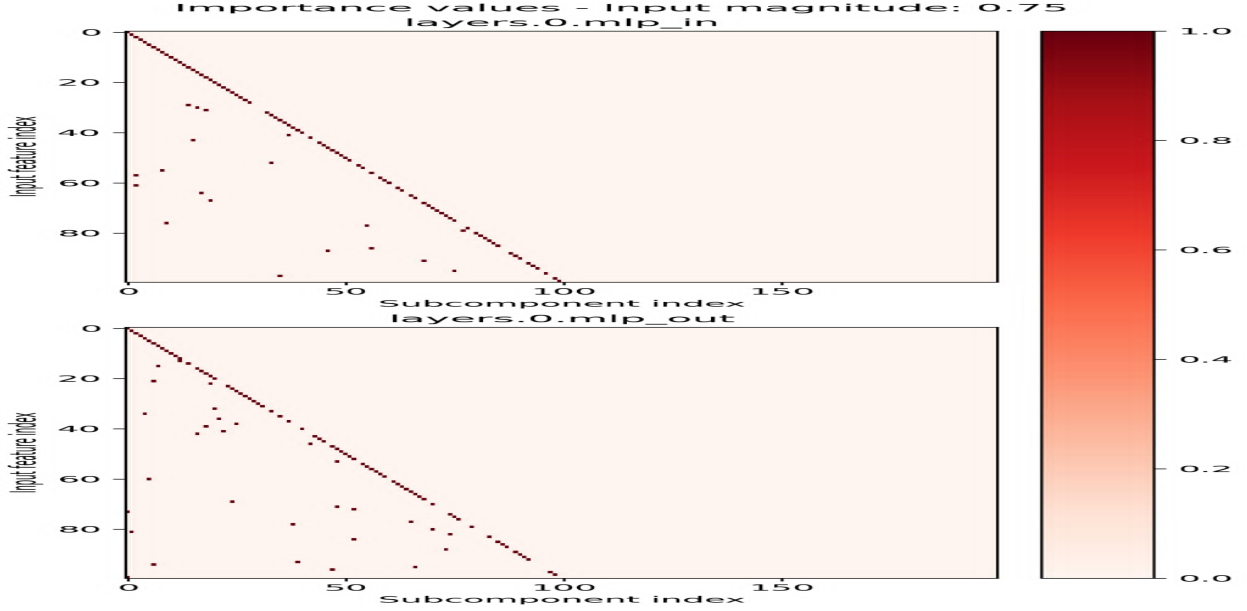


Figure 5: Causal importance values for Softmax on the Toy Model of Compressed Computation. Each input feature activates a near one-hot importance pattern across subcomponents. Sampled using one-hot sampling.

If this reading is taken at face value, there should only be two possible interpretations, either each subcomponent learns a monosemantic representation for one feature, and the remaining subcomponents are unimportant and can be ablated. Alternatively, the minimality loss is too strong and forces the model to concentrate all importance on a single subcomponent, at the cost of reconstruction which should be reflected in a high reconstruction error. Surprisingly, as seen in table 4, Softmax produces near-zero reconstruction error, and should therefore have recovered a monosemantic decomposition.

To examine this, the neuron contributions for both the target model and softmax are plotted in figure 6. The computation of the contribution is computed as in SPD [BBS25].

$$\text{NeuronContribution}_{\text{model}}(i) = (W_{U[i,:]}W_{out}) \odot (W_{in}W_E[:,i]) \quad (18)$$

$$\text{NeuronContribution}_{\text{subcomponent}}(i) = \max_{c \in C} [(W_{U[i,:]}U_{out}V_{out}^T) \odot (U_{in[:,c]}V_{in}^T[c,:]W_E[:,i])] \quad (19)$$

where $W_{E[:,i]}$ and $W_{U[i,:]}$ are the i -th column and row of the embedding and unembedding matrices and $\odot$ denotes elementwise multiplication.

$W_{U[i,:]}W_{out}$ measures how strongly the neurons contribute to the "reconstruction" of the output, while $W_{in}W_E[:,i]$ measures how strongly the neurons are activated by the input feature i . If both produce high values, the product will be high, indicating that the neuron contributes strongly to the feature's computation. Otherwise if either is low, the contribution will be low, indicating that the neuron does not contribute significantly to that feature.

As seen in figure 6, Softmax, like SPD, find subcomponents that contribute to multiple features, which also replicates the behaviour of the target model.

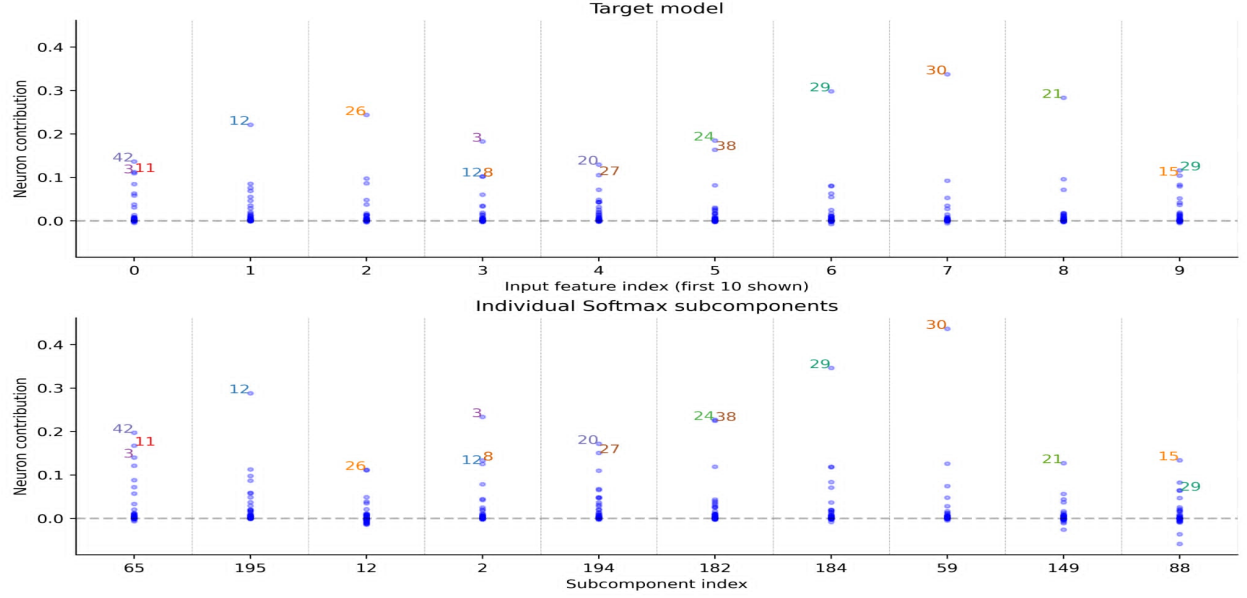


Figure 6: Per-feature contribution of each subcomponent at the neuron level. Subcomponents contribute to multiple features simultaneously, even when their causal importance is concentrated on a single feature.

The one-hot importance assignment is therefore not equivalent to a monosemantic parameter decomposition. One initial reading would be that the subcomponents represent multiple features in superposition, with the observed interference arising from near-orthogonal subcomponents. However, this is contradictory to the results found by [Bha+25], which show that the trained W_{out} operates within a shared 50-dimensional subspace rather than implementing 100 separate computations.

If softmax was genuinely encoding 100 features in superposition, the U_c and V_c vectors of its 100 active subcomponents would have to be near-orthogonal. To examine this, the pairwise cosine similarity between the U_c and V_c vectors of the causally important subcomponents are computed in figure 7.

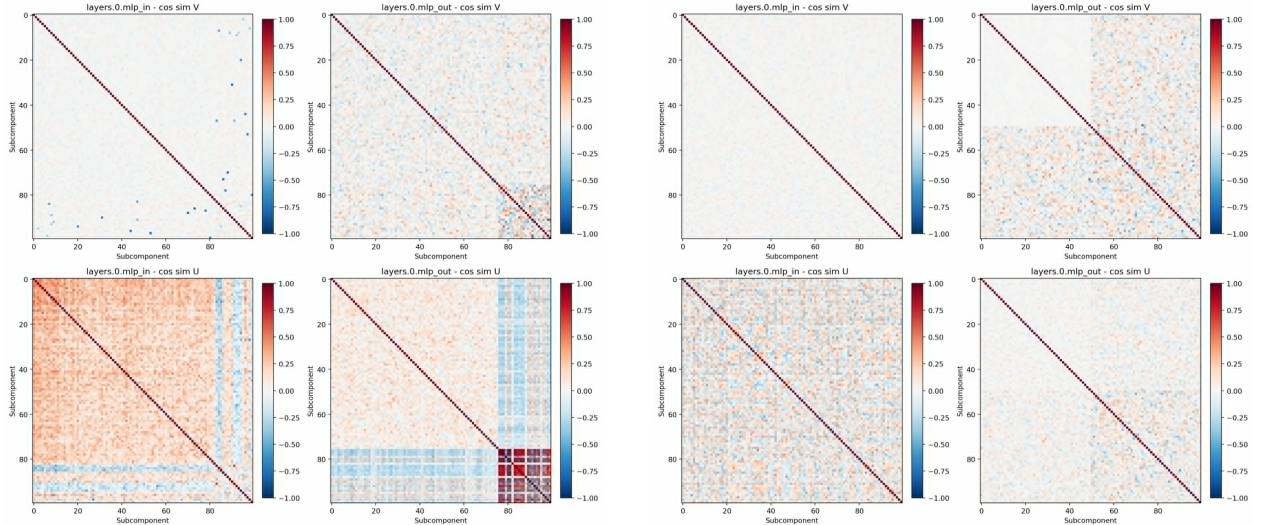


Figure 7: Left: Softmax. Right: SPD. Pairwise cosine similarity between the U_c and V_c vectors of the causally important subcomponents for both layers. Subcomponents are ordered by decreasing norm.

For W_{in} , both methods recover near-orthogonal V vectors, an expected result, as the embedding dimension is large enough to accommodate all pairwise orthogonal input features.

The U vectors, by contrast, must project into the 50-dimensional MLP hidden space and therefore cannot be pairwise orthogonal. Interestingly, the two methods find wildly different projections in the U space. Softmax finds two clusters, where the first 80 subcomponents have large positive cosine similarities with each other and large negative similarities with the remaining 20, while the final 20 are positively correlated among themselves. SPD on the other hand find many subcomponents sharing similar directions, but without the clear two-cluster structure observed in SPD.

For W_{out} , the two methods diverge in their representation. In SPD, the 50 causally important subcomponents recover orthogonal directions in both U and V , while the remaining subcomponents are entangled but carry negligible norm and contribute nothing to the final projection.

The softmax variant, which assigns causal importance to 100 subcomponents, presents a more entangled picture: the first 80 subcomponents have small pairwise cosine similarities in V , while the remaining 20 collapse onto highly similar directions. The U vectors on the otherhand, find the same two-cluster pattern as in W_{in} , with the first 80 subcomponents having small positive cosine similarities with each other and large negative similarities with the remaining 20, while the final 20 practically collapse onto the same direction.

4.4 Subcomponent Norm Distributions

As briefly mentioned in TMS, but also observed in the CC model, softmax produces feature shrinkage, even when it produces the right feature directions. To quantify if this could be the cause of insufficient norm concentration, the norms of the subcomponents are examined. (Figure 8)

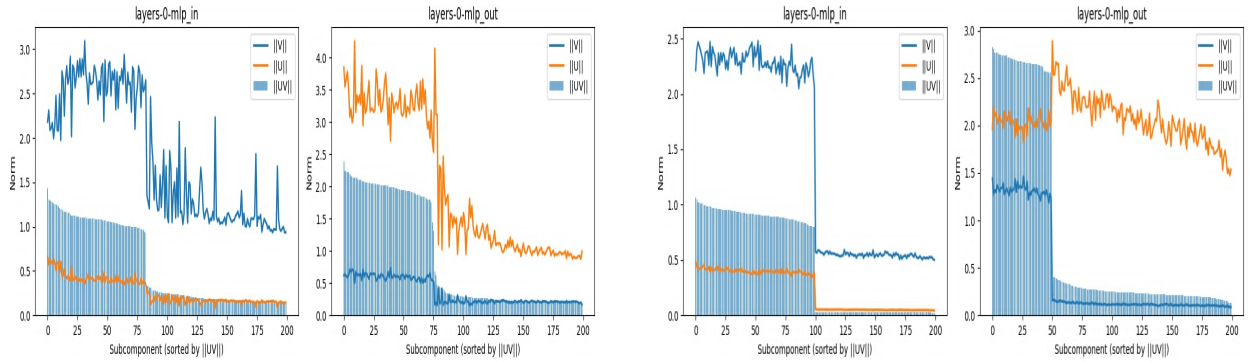


Figure 8: Left: Softmax. Right: SPD. Norms of the subcomponents, ordered by decreasing norm. Matrix norm is the fribenius norm of the subcomponent matrix $U_c V_c^T$. Subcomponents are ordered by decreasing norm.

The two methods recover very different norm distributions. Consistent with the cosine similarity analysis, softmax separates into two broad clusters: approximately the first 80 subcomponents have substantially larger norms, while the remaining 120 retain smaller but non-negligible norms in both W_{in} and W_{out} .

Notably, the softmax does not drive unused subcomponents to zero norm, but rather to a small but non-negligible value.

SPD, by contrast, concentrates norm almost exclusively in the causally important subcomponents: approximately 100 subcomponents for W_{in} and 50 for W_{out} . The remaining subcomponents are effectively pruned, with norms close to zero rather than forming the residual low-norm tail observed in the softmax decomposition.

A further observation is that the norm contributions differ systematically between the two layers. In W_{in} , the norm is dominated primarily by the V vectors, whereas in W_{out} the dominant contribution instead comes from the U vectors. This pattern is consistent across both decomposition methods.

An initial hypothesis for why the softmax formulation fails to produce hard-zero norms was that, unlike the hard-sigmoid gating used in SPD, the softmax distribution cannot produce exact zeros. However, the plot over the causal importance value frequencies rule out this explanation (Figure 9). The learned importance assignments are nearly bimodal, with most concentrated at either 0 or 1, and only a small fraction occupying intermediate values. (Figure 9) The absence of hard-zero norms therefore seems not to arise from the gating distribution itself, but rather from how reconstruction mass is distributed across subcomponents during optimisation. Empirically, this hypothesis was further weakened by replacing the softmax gating with sparsemax [PNM19], which is capable of producing exact zeros in

the gating distribution. Despite this, the resulting norm distributions remained qualitatively identical to those obtained with softmax.

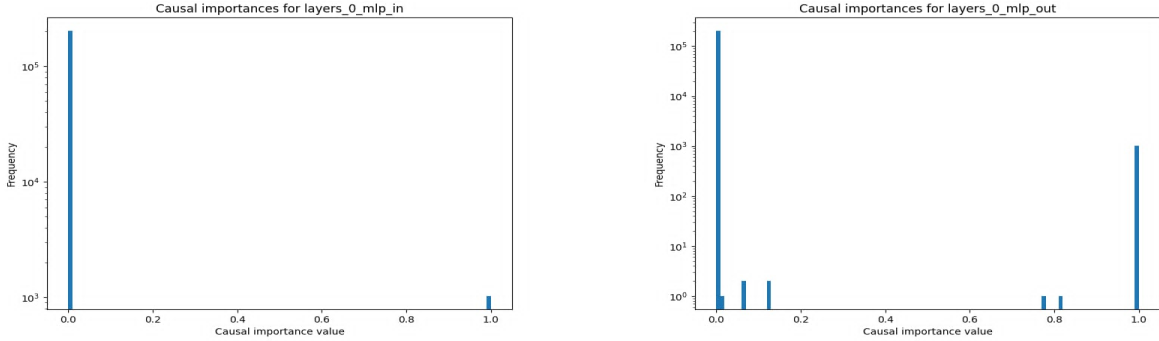


Figure 9: Distributions of causal importance values for the softmax decomposition on both layers.

More generally, this behaviour is likely related to the feature shrinkage phenomenon previously observed in both TMS and CC. The faithfulness objective is computed additively as the MSE between the summed subcomponent reconstruction and the target weight matrix, making it indifferent to how reconstruction mass is allocated across subcomponents. Consequently, if the importance regularisation only weakly penalises low-magnitude subcomponents, optimisation may converge to solutions in which a small set of dominant subcomponents carry most of the semantic structure, while a long tail of weakly active components absorbs residual reconstruction error.

4.4.1 Noes for geometry

To quantify this, the effective rank of each pair’s submatrix, $W_{:,2i-1:2i} \in \mathbb{R}^{d_{\text{hidden}} \times 2}$ is computed using the entropy-based definition of [RV07].

$$r_{\text{eff}}(W_g) = \exp\left(-\sum_{k=1}^Q p_k \log p_k\right), \quad p_k = \frac{\sigma_k^2}{\sum_{j=1}^Q \sigma_j^2} \quad (20)$$

where σ_k are the singular values of W_g and $Q = |g|$. The quantity lies in $[1, |g|]$ and equals 1 when the group is exactly rank-one. To gauge the required value of k for the rank- k extension, the maximum effective rank across pairs is examined:

$$k = \max_i r_{\text{eff}}(W_{:,2i-1:2i}), \quad (21)$$

In this case the max is essentially 1 across all pairs, suggesting that $k = 1$ is sufficient to capture these features without loss of information. (Table 5).

Mean	Std Dev	Range
1.0003	0.01	[1.0, 1.05]

Table 5: Effective rank of paired-feature encoder blocks. Values near 1 indicate collapse to a single direction.

Related to the table of experiments, the whole point is that there is no ground truth decomposition consisting only of the sides like the cube example !

Remarks for this section Intra cluster cosine has a -1. Two features are perfectly aligned to a factor of -1, can this cause issues with decomposition? Same component could capture both features, if they read from the same input. However they shouldnt as its distinctly different feature indices.

4.5 Toy model of geometry

In the paired TMS regime, even though input features were nominally two-dimensional, the model collapsed them onto a single direction, so a rank-1 decomposition per feature still suffices. To test the rank- k extension of Section 3.3,

a setting is needed where features cannot collapse to a single direction without losing essential structure. This experiment generalizes the cube example from the introduction, where the same underlying computation admits multiple valid ground-truth decompositions depending on what each subcomponent is allowed to represent.

In this experiment, we train a model to compute the volume of an n -dimensional simplex from its $n + 1$ vertices in $\mathbb{R}^n$. The model should learn to compute the volume as follows:

$$\text{Vol}(v_1, \dots, v_{n+1}) = \frac{1}{n!} |\det([v_2 - v_1 \quad v_3 - v_1 \quad \dots \quad v_{n+1} - v_1])|. \quad (22)$$

The computation necessarily involves combining all vertices together, and would functionally only make sense, if the model captures the joint structure of the vertices rather than treating them as independent features.

Perhaps something about affine independence?.

To mirror the paired-feature setup of Section 4.2.2, the model receives K simplices as input, where each simplex is either sampled with probability p or all of its coordinates set to zero otherwise.

$$f(x) = W_1 \text{ReLU}(W_2 x), \quad W_2 \in \mathbb{R}^{d_{\text{hidden}} \times K n(n+1)}, \quad W_1 \in \mathbb{R}^{K \times d_{\text{hidden}}}. \quad (23)$$

here $n(n + 1)$ represents the total number of coordinate points a simplex has in $\mathbb{R}^n$, and K is the number of simplices. Unlike TMS, d_{hidden} is chosen large enough that the model does not need to compress features through superposition. This should ensure that the model can learn to assign a subset of its neurons to each simplex, giving a clearer ground truth decomposition. Concretely, for this experiment, the simplices are triangles in $\mathbb{R}^2$. Secondly, we set $K = 3$ and $d_{\text{hidden}} = 150$, making the input space $\mathbb{R}^{18}$.

Intuitively it seems reasonable, that a decomposition could recover a coordinate, a vertex, a side or the whole simplex. Which decomposition should be recovered depends on 1) the dimensionality that each subcomponent is allowed to represent, and secondly how strongly the importance function penalises spreading importance across multiple subcomponents. Particularly, for softmax sampling with entire simplices at a time, the entropy minimality maximizes the probability of a simplex belonging to a single subcomponent, so we would expect, that given the right dimensionality and sufficient pressure, that the decomposition would recover the whole simplex as a single component, rather than splitting it across multiple components.

To isolate the effect of these two factors, will only vary one of them at a time, while keeping the other fixed. Particularly, the rank will be fixated at $k = 1$, which will recover the original SPD method, and examine how the minimality coefficient β_3 influences the decomposition. Then, the minimality coefficient will be fixated at a value that produced a clear decomposition, at $k = 1$ and examine how increasing the rank allows for different decompositions to be recovered.

4.6 Toy model of geometry

In the paired TMS regime, even though input features were nominally two-dimensional, the model collapsed them onto a single direction, so a rank-1 decomposition per feature still suffices. To test the rank- k extension of Section 3.3, a setting is needed where features cannot collapse to a single direction without losing essential structure. This experiment generalizes the cube example from the introduction, where the same underlying computation admits multiple valid ground-truth decompositions depending on what each subcomponent is allowed to represent.

The model is trained to compute the volume of an n -dimensional simplex from its $n + 1$ vertices in $\mathbb{R}^n$:

$$\text{Vol}(v_1, \dots, v_{n+1}) = \frac{1}{n!} |\det([v_2 - v_1 \quad v_3 - v_1 \quad \dots \quad v_{n+1} - v_1])|. \quad (24)$$

By construction, the $n + 1$ vertices of a non-degenerate simplex are affinely independent: no vertex lies in the affine hull of the others. The volume is therefore a genuine function of all vertices jointly, and no subset can be dropped without changing it. This makes the simplex an irreducible feature in the sense of Section 3.2, with internal dimensionality genuinely greater than one.

To mirror the paired-feature setup of Section 4.2.2, the model receives K simplices as input, each sampled with probability p and otherwise set to zero across all of its coordinates:

$$f(x) = W_1 \text{ReLU}(W_2 x), \quad W_2 \in \mathbb{R}^{d_{\text{hidden}} \times K n(n+1)}, \quad W_1 \in \mathbb{R}^{K \times d_{\text{hidden}}}, \quad (25)$$

where $n(n + 1)$ is the total number of coordinates of a simplex in $\mathbb{R}^n$ and K is the number of simplices. Unlike TMS, d_{hidden} is chosen large enough that the model does not need to compress features through superposition, which should

allow it to assign a separate subset of neurons to each simplex and produce a clearer ground-truth decomposition. Concretely, the simplices are triangles in $\mathbb{R}^2$, with $K = 3$ and $d_{\text{hidden}} = 150$, giving an input space of $\mathbb{R}^{18}$.

A decomposition could in principle recover a single coordinate, a vertex, an edge, or an entire simplex. Which of these is recovered depends on two factors: the dimensionality each subcomponent is allowed to represent, and how strongly the importance function penalises spreading importance across multiple subcomponents. For the softmax variant in particular, where entire simplices are sampled together, the entropy minimality maximises the probability of a simplex belonging to a single subcomponent. Given sufficient rank and sufficient minimality pressure, the decomposition is then expected to recover a whole simplex as a single component rather than splitting it across several.

To isolate the effect of these two factors, only one is varied at a time. First, the rank is fixed at $k = 1$, recovering the original SPD formulation, and the minimality coefficient β_3 is varied to examine its influence on the decomposition. Second, β_3 is fixed at a value that produced a clear decomposition at $k = 1$, and the rank is increased to examine how higher k allows different decompositions to be recovered.

Target model structure. Before examining what the decomposition methods recover, it is worth confirming the structure of the target model itself, since a clean ground-truth decomposition exists only if the model processes each simplex through a separable set of neurons.

The input is partitioned into K groups, where group g contains the $n(n+1)$ coordinates of simplex g , indexed by the set $\mathcal{C}_g$. The preactivation of hidden neuron i decomposes additively over these groups,

$$\sum_c W_{2[i,c]} = \sum_{g=1}^K \rho_{i,g}, \quad \rho_{i,g} := \sum_{c \in \mathcal{C}_g} W_{2[i,c]}, \quad (26)$$

where $\rho_{i,g}$ is the influence of simplex g on neuron i . These per-simplex influences are plotted across all neurons and groups in the top panel of Figure 10.

On the output side, the contribution of neuron i to the volume output of simplex k is read directly from $W_1 \in \mathbb{R}^{K \times d_{\text{hidden}}}$ as $W_{1[k,i]}$ (bottom panel of Figure 10).

Lastly, to quantify which input-to-output paths are functionally relevant, only each neuron’s dominant input and dominant output are considered. For neuron i , let

$$g^*(i) = \arg \max_g \rho_{i,g}, \quad k^*(i) = \arg \max_k W_{1[k,i]} \quad (27)$$

be the simplex that most strongly drives the neuron and the output to which it most strongly contributes computing the inner product of these two values gives a measure of, how relevant the strongest input-output path through that neuron is. These are displayed in the top panel of Figure 10.

. Accumulating the product of these maximal values over all neurons gives the group-to-output matrix

$$M_{g,k} = \sum_{\substack{i : g^*(i)=g \\ k^*(i)=k}} \rho_{i,g} W_{1[k,i]}, \quad (28)$$

where each neuron contributes only to the cell defined by its dominant input and output. A large $M_{g,k}$ indicates that many high-magnitude neurons read primarily from simplex g and write primarily to output k . The resulting $K \times K$ matrix is shown in Figure 11, whose diagonal structure confirms that each simplex routes to its own output.

Sorting the neurons by their dominant input group and dominant output reveals a clear block structure in both panels: each input group activates a disjoint set of neurons, and each of those neurons contributes to a single output. Sampling complete simplices and measuring the resulting output collapses this into a $K \times K$ group-to-output response (Figure 11), which is diagonal: simplex g drives neurons that contribute only to output g .

This confirms that the K simplices are processed by K non-overlapping subnetworks. Because subcomponents combine additively to reconstruct the model, and each simplex occupies a disjoint subnetwork, there exists a decomposition in which each subcomponent captures the entire contribution of exactly one simplex, provided the rank is sufficiently large. Recovering this decomposition would make the post-hoc clustering trivial, with one subcomponent mapping to one simplex.

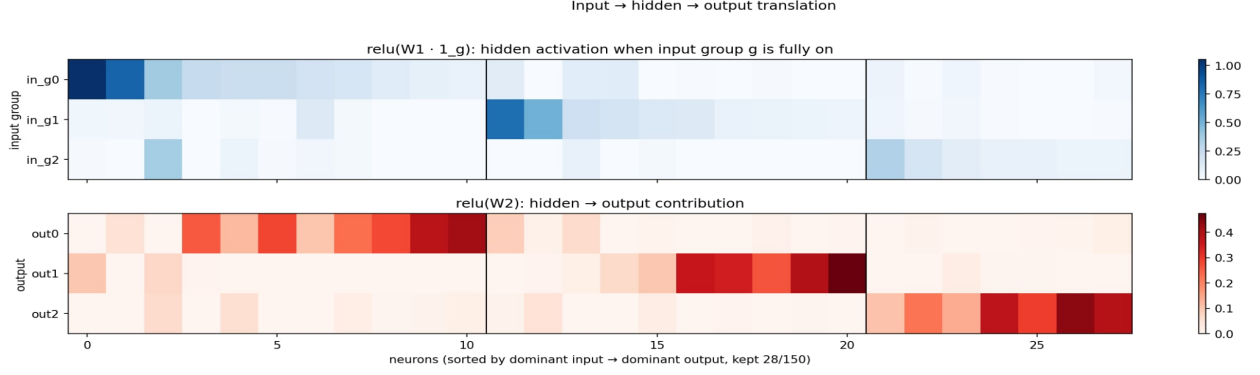


Figure 10: Top: response $\rho_{i,g}$ of each hidden neuron to each input group, when group g is activated in isolation. Bottom: contribution $W_{1[k,i]}$ of each hidden neuron to each simplex’s volume output. Neurons are sorted by dominant input group then dominant output, with only the 28 active neurons of $d_{\text{hidden}} = 150$ shown. Both panels show that each simplex occupies a disjoint set of neurons.

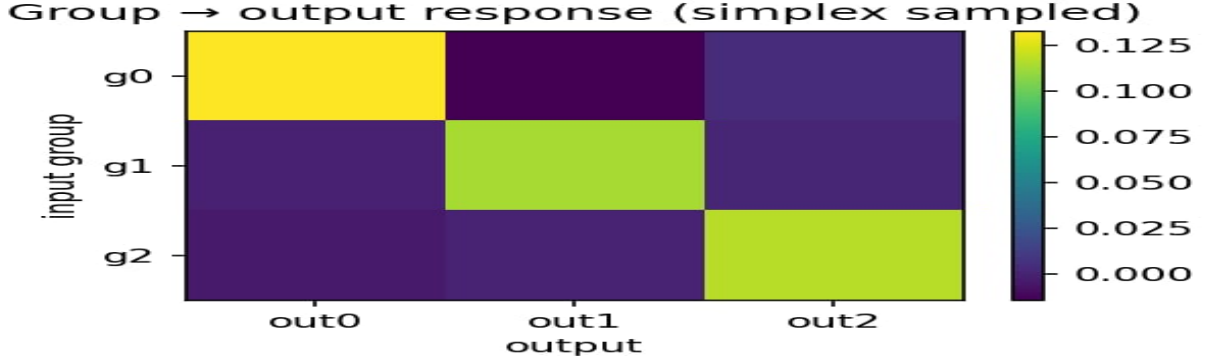


Figure 11: Group-to-output response obtained by sampling complete simplices and measuring the volume output. The diagonal structure confirms that simplex g contributes only to output g , with no cross-simplex interaction.

4.7 Comparison of importance values

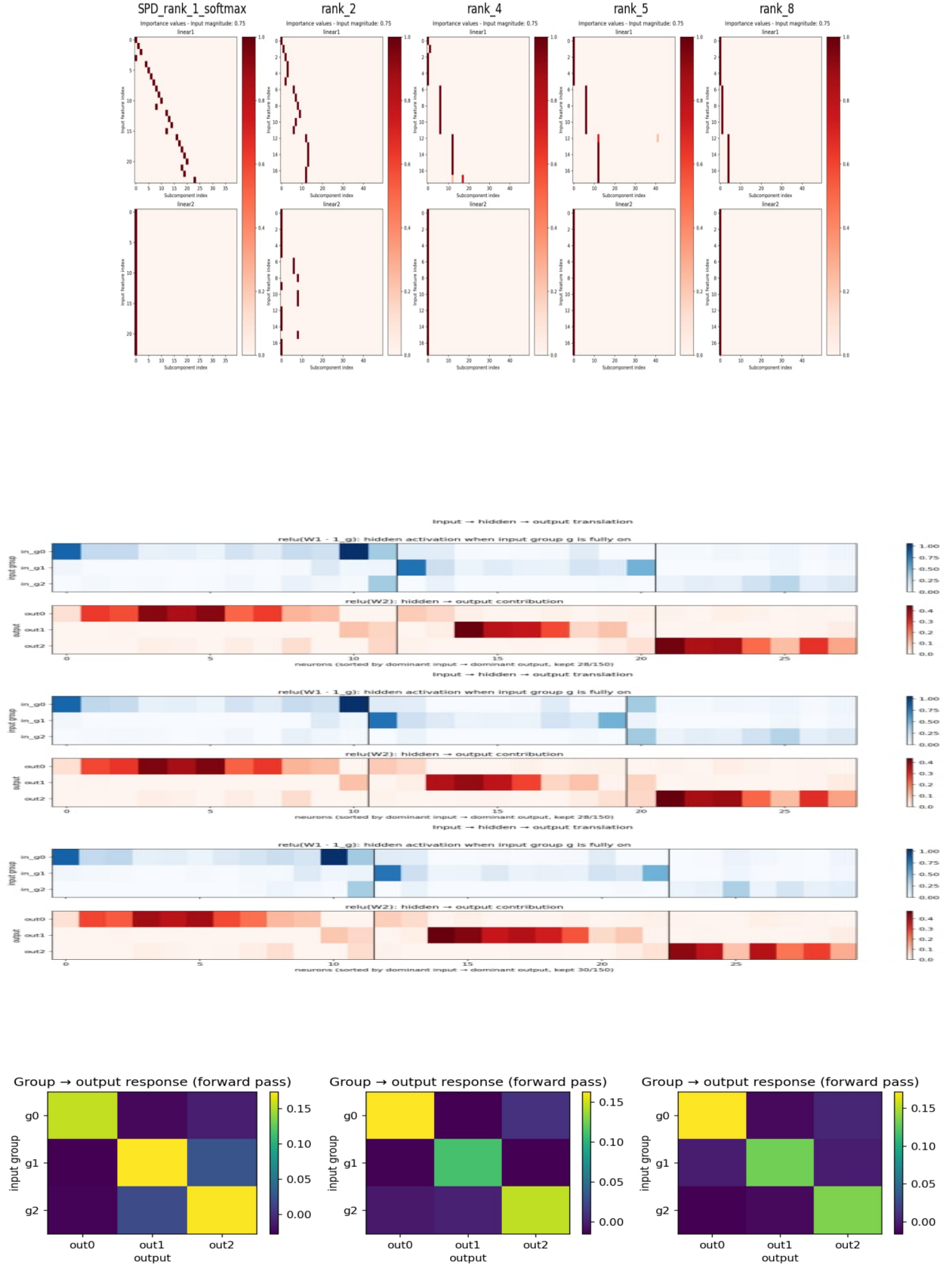
The importance values for

4.7.1 Comparison of importance values across different minimality coefficients

Mention how losses change too!

4.7.2 Comparison of importance values across different ranks for fixed minimality.

mention if it affects losses.



4.8 Computing

Ideally after constructing these rank k components, then it should be feasible to fully reconstruct the output of the model, by summing these causally important subcomponents together, without any loss of information. In this case the rank k component should fully capture it.

We try this only for Layer 1 and layer 2 first,
see if they replicate the layer output.

We do the input, output plot, where we only probe the causally important subcomponents.

5 Discussion

The original spd method produces hard zeroes, something that softmax cannot do. Entmax didnt work either nor did sparsemax, however non hard zeroes do cause magnitude to leak even if marginal, which may increase with more components.. important to mention that the structure that spd decomposes into is not clear sometimes one hot some.

Talk about how its unclear what structure spd wants. Even rank 1 is not the smallest possible, as we can align two things to the same direction, but these do not split.

Relate back to the interesting thing that one feature per component and rank 1 matrices does in fact not mean the smallest possible decomposition, as we have shown with TMSs and TMS paired, one is in a way strictly larger, but they result in the same thing!

Talk about how softmax if working correctly, is probably better at extracting relevant features, not how it decomposes of smaller parts. Here it could be nice to touch on, when there genuinely is a one hot solution contra not, i.e in a lower dimensional subspace.

Thesis needs to discuss how many of the scenarios from the irreducibility section are actually tested -Tms serves as dependent features that cannot be split, -CC serves as an independent feature - idk if we can do something with wout in this sense? - Quantization, needs to somehow engage with the two approximation methods, that irreducibility presents. -Here we can then say if we have a mixture or if not -Most likely conclusion becomes that we only discuss cases, where the ground truth is dependence not if it manages to split non irreducible features.

Discussion needs to engage with the point of, should the importance value try to replicate model behaviour or should it try to find a more disentangled representation like SAE's do?

6 Acknowledgements

7 Other things to consider?

Minimal descriptive length?

7.1 That some features are multidimensional

Think back to SAE, that learn a dictionary of features, then find the linear combination of those features, however that this fails for cyclical features, as its cheaper to learn a single feature, than learn a direction and a periodic function. Now for your experiment, the dream scenario is that we allow a multi dimensional component, and that this can learn the cyclical feature, i.e a weekly cycle or modular arithmetic either do this or extend TMS to be multi-dimensional, show if it works better with ur gnn or not, likewise we need to show that through individual masking of features, we can teach a component to kill one of its features, if it is better explained by a rank 1 component than rank 2.

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9 Appendix

Descriptive length thing? + approximation of attention experiment

10 Notes

Putting the minimality Isos entropy higher, encourages diagonals? Instead of capturing two features in the same component.

Increasing minimality loss encourages feature grouping.

SAE: learn to decompose activations, i.e the output of a layer, Transcoders: learn to approximate the input-output function of a layer

$$\int p(a, b) da db = w \int p_A(a) da \int \delta(b) db + (1 - w) \iint p_{AB}(a, b) da db = w + (1 - w) = 1$$

to read: Terence Tao, *An Introduction to Measure Theory*: Section 1.1 Prologue: The problem of measure — sets up why measure theory exists and why ”lower-dimensional sets have measure zero” is the central idea. This alone might give you the intuition you need. Section 1.2 Lebesgue measure — gives you ”measure zero,” ”Lebesgue measure on $\mathbb{R}^n$,” and the fact that lines/planes/etc. have measure zero in higher-dimensional ambient spaces. This is the technical core for your question. Section 2.3 Probability spaces — short section that frames everything in probability language, which is what you actually want.

10.0.1 Hard maybes

10.0.2 Per-dimension ablation and implicit rank selection (Hypothesis: ask lukas)

Following the extension to rank- k subcomponents, and the alternative formulation of the GNAN, the output $g_c^l \in [0, 1]^k$ is a per feature importance value, and thus the masking can be applied independently to each feature. The intuition behind this is that if a mechanism captures a 1 dimensional feature, it would only need of the k dimensions to be active.

Likewise if this actually ends up working, k can be considered an upper bound or a so called ”capacity” of a mechanism, it also follows nicely with the descriptive length principle found in the appendix of APD.

To determine the effective rank of each subcomponent, the stochastic masking becomes:

$$[m_c^l(x, r)]_j = [g_c^l(x)]_j + (1 - [g_c^l(x)]_j) r_{c,j}^l, \quad r_{c,j}^l \sim U(0, 1) \quad (29)$$

for each j in $1, \dots, k$

Notably this treats each feature independently, which may not be desirable, as such we could modify it ever so slightly:

$$[m_c^l(x, r)]_j = [g_c^l(x)][g_c^l(x)]_j + (1 - [g_c^l(x)][g_c^l(x)]_j) r_{c,j}^l, \quad r_{c,j}^l \sim U(0, 1) \quad (30)$$

The signal becomes an importance of two factors, if the component is important and both features are important, it remains active, if only one feature is important, the other will be pushed to zero

The masked weight matrix becomes:

$$W^l(x, r) = \sum_{c=1}^C U_c^l \text{diag}(\mathbf{m}_c^l(x, r)) V_c^{l\top} \quad (31)$$

Where the diagonal matrix is the per-feature mask for subcomponent c .

The importance minimality loss penalises all k dimensions of all subcomponents toward zero:

$$\mathcal{L}_{\text{import}} = \sum_{l=1}^L \sum_{c=1}^C \sum_{j=1}^k |[g_c^l(x)]_j|^p \quad (32)$$

This follows the same logic by which SPD selects the number of active subcomponents: the minimality loss drives unused components toward zero, and only those that are genuinely required to reconstruct the target model’s output survive.

This makes k a soft upper bound rather than a sensitive hyperparameter. In practice, k can be set conservatively large and the effective rank of each subcomponent is learned from the data.

11 To add:

<https://arxiv.org/pdf/2303.13506> quantization, multidimensional feature crossover We use the method from the irreducibility paper to quantize how well it is seperable, mixture like. Rewrite irreducibility to mkae it clear that it is the function

The method is quite complex, we practically look for the input space to see where a feature comes from and its dimension but when we look to decompose a layer, we look at f , the projection of the input space to see its rank, but we look for the values that the input space get mapped to. So in other words, the f decides how many subcomponents we need, but the f decides how many features are inside of it?

quantization and spd clustering

why softmax + attention instead of only softmax + minimality.

Somewhere add this whole thing of importance maybe being an interpretability trap? Or that it should be seen as a caution thing, it will predict what is responsible, not the model alignment.

The thing that makes multifeature hypothesis perfect

Multifeature hypothesis says that there exists a transformation f , to a lower dimensional space, in this case any W linear map,

that can be paired with some new V_i that describes the hidden state, in this case it’s the SPD components!!!! These perfectly make this up and pair this wiht quantization and we are golden.

Remark rewrite TMS paired into TMS, they should be the same section. Rme Remember V is the input reading so if w

In general add how each experiment relates to the multi dimensional feature hypothesis decomposition.

11.0.1 SVD reformulation, capacity maybe per feature ablation TBD

+ add read write streams, theyre a general thing - ζ perhaps issues with diffusion and recurrent models where this stream is not as clear. <https://arxiv.org/pdf/2504.00194> + This section must explain delta orthogonality, visualization of Read, write matrices, perhaps nudge the SVD view. Most importantly should address if models learn lower dimensional representations or if its another k sensitivity as in APD.

The competition pressure introduced by the softmax has a theoretical motivation under the multi-dimensional superposition hypothesis. The hypothesis predicts that different irreducible features occupy approximately orthogonal projection subspaces; the softmax-and-attention mechanism implements a training pressure toward this orthogonality at the level of SPD’s atoms. Atoms whose V matrices share input directions cannot be cleanly distinguished on inputs that activate those directions, and the softmax penalises ambiguous allocations. The optimiser’s response is to push the atoms’ V matrices toward disjoint subspaces, recovering the orthogonality structure the hypothesis posits.

Remember to add an experiment that shows if softmax enforce the orthogonality constraint from engels.

12 Lukas questions

- Show him CC, - Show him geom - ζ ideas are that i want to both show how ”unimportant” the importance values are, for the final decomposition. no readout is still faithful - Ask about irreducibility - ζ i define mixtures and so fourth, but

its not really used anywhere. Is that okay? It's really hard to assess if it will be one or not. - It iwll be in my discussion that we don't examine the mixture case directly. "The independence" is technically between input pairs, and within pair is dependent. - Ask about formalities, should i do a quey over how ive done litterature search, should my appendix contain my hyperparameter strategy, or can i just link wandb reports, with hyperlinks? - Do you want the github repo attached to my handin? - I will exceed the page limit if i use this many images, but thats ok right?

-wants code

-wandb + hyperparameters in appendix.